

business sharks Lukas and Julien

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#Introduction

This study focuses on the excess returns generated by American politicians' stock trades. The main objective is to analyze the relationship between excess returns and net gains across political party, state, and individual politicians.

1) Organization of the data

##	[1]	"Ticker"	"TickerType"	"Company"
##	[4]	"Traded"	"Transaction"	"Trade_Size_USD"
##	[7]	"Status"	"Subholding"	"Description"
##	[10]	"Name"	"Filed"	"Party"
##	[13]	"District"	"Chamber"	"Comments"
##	[16]	"Quiver_Upload_Time"	"excess_return"	"State"
##	[19]	"last_modified"		

The dataset consists of 17 columns, as presented above. However, the raw data required reorganization and cleaning before analysis.

After preprocessing, several unnecessary columns were removed and missing column names were eliminated. Date variables were standardized across all observations. Trade size values reported as ranges were converted into minimum and maximum values, and a midpoint measure was constructed.

Additional variables were created, including:

Trade delay, measured in days between the trade date and filing date

Net trade direction, where purchases were coded as +1 and sales as -1

Midpoint trade size, representing the average of the reported range

Net gain, calculated as excess return multiplied by trade direction

##	[1]	"Ticker"	"Company"	"Traded"
##	[4]	"Transaction"	"Trade_Size_USD"	"Subholding"
##	[7]	"Description"	"Name"	"Filed"
##	[10]	"Party"	"District"	"Chamber"
##	[13]	"Comments"	"excess_return"	"State"
##	[16]	"last_modified"	"Trade_Size_USD_min"	"Trade_Size_USD_max"
##	[19]	"trade_delay"	"net_trade_direction"	"trade_size_mid"
##	[22]	"net_gain"		

After reorganization, the dataset now contains 21 columns. Three original variables were removed and seven new variables were added. The most important constructed variables are net gain, midpoint trade size, and trade delay. Net gain reflects the economic outcome of each transaction by combining excess return with trade direction.

2) Trade Size and number

We examine both the size and the number of trades on a yearly and monthly basis and compute correlations between these measures.

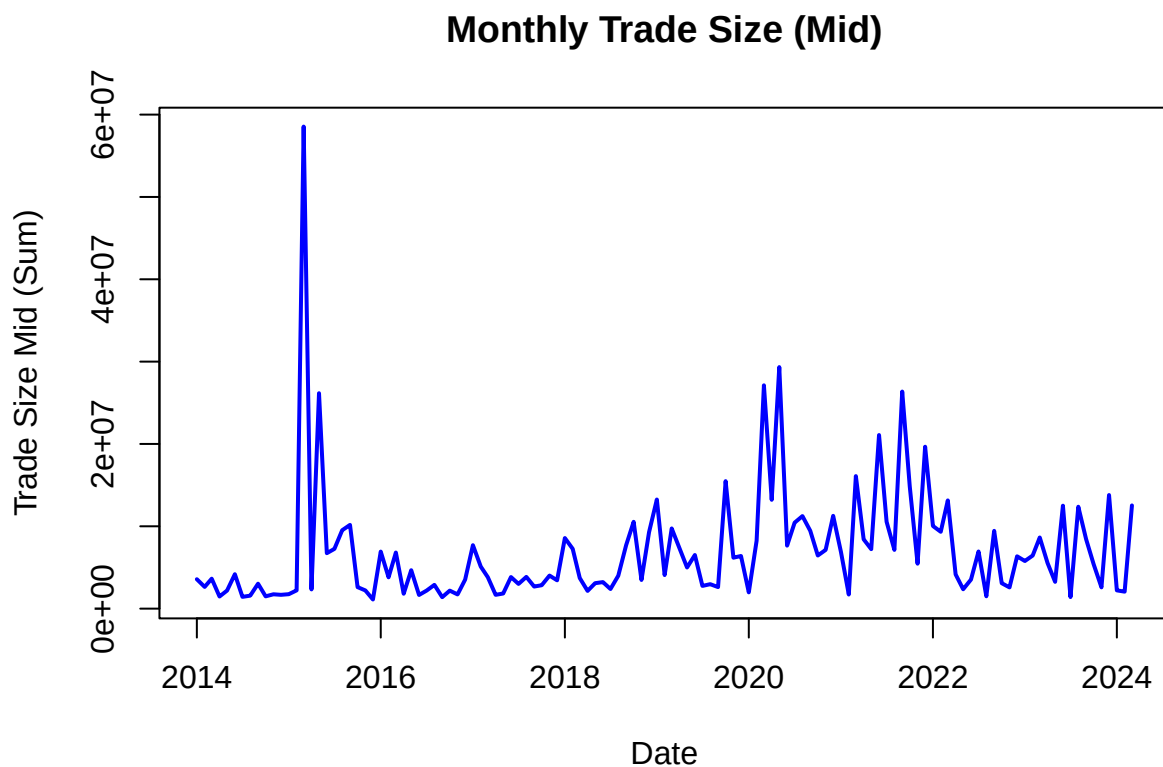
The mid size of trades per year is presented as follows:

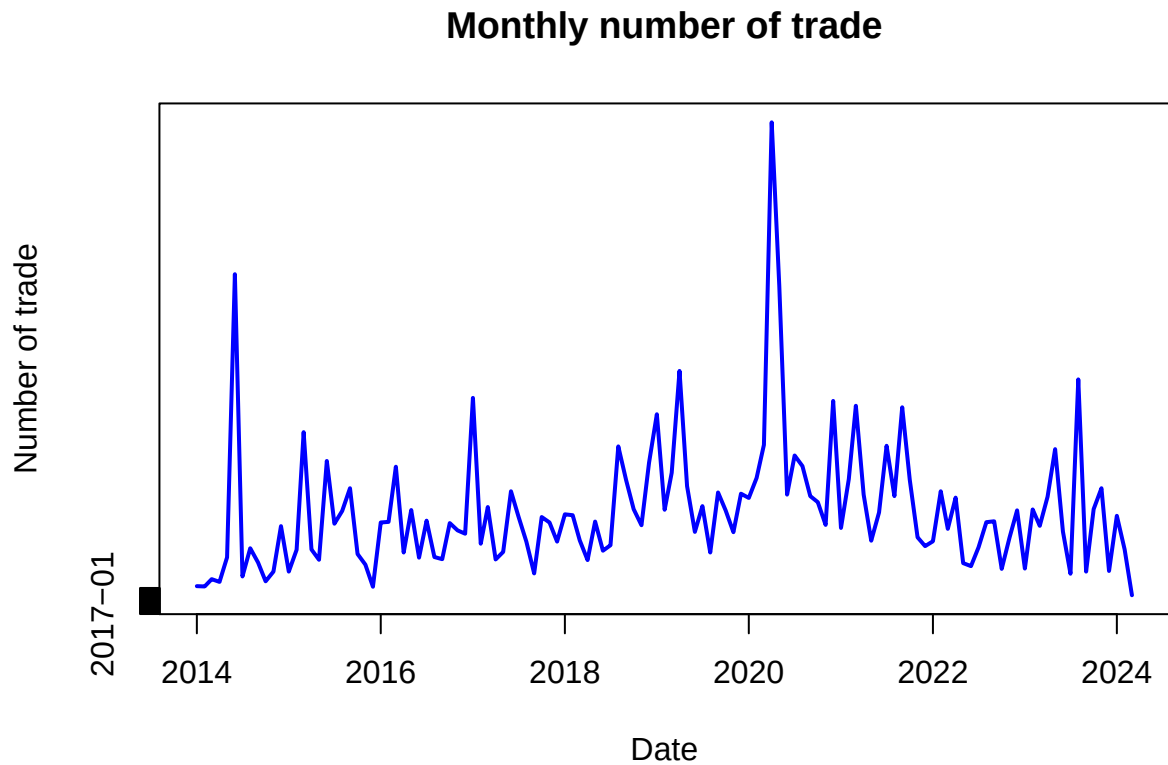
##	2014	2015	2016	2017	2018	2019	2020	2021
##	66154103	228706399	87088054	98730360	150494902	180750399	349174081	328230682
##	2022	2023	2024					
##	170013831	206920068	45168016					

While the number of trades per year is:

##	years											
##	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	
##	3205	3797	3786	4087	4394	5377	7699	5377	3652	4383	710	

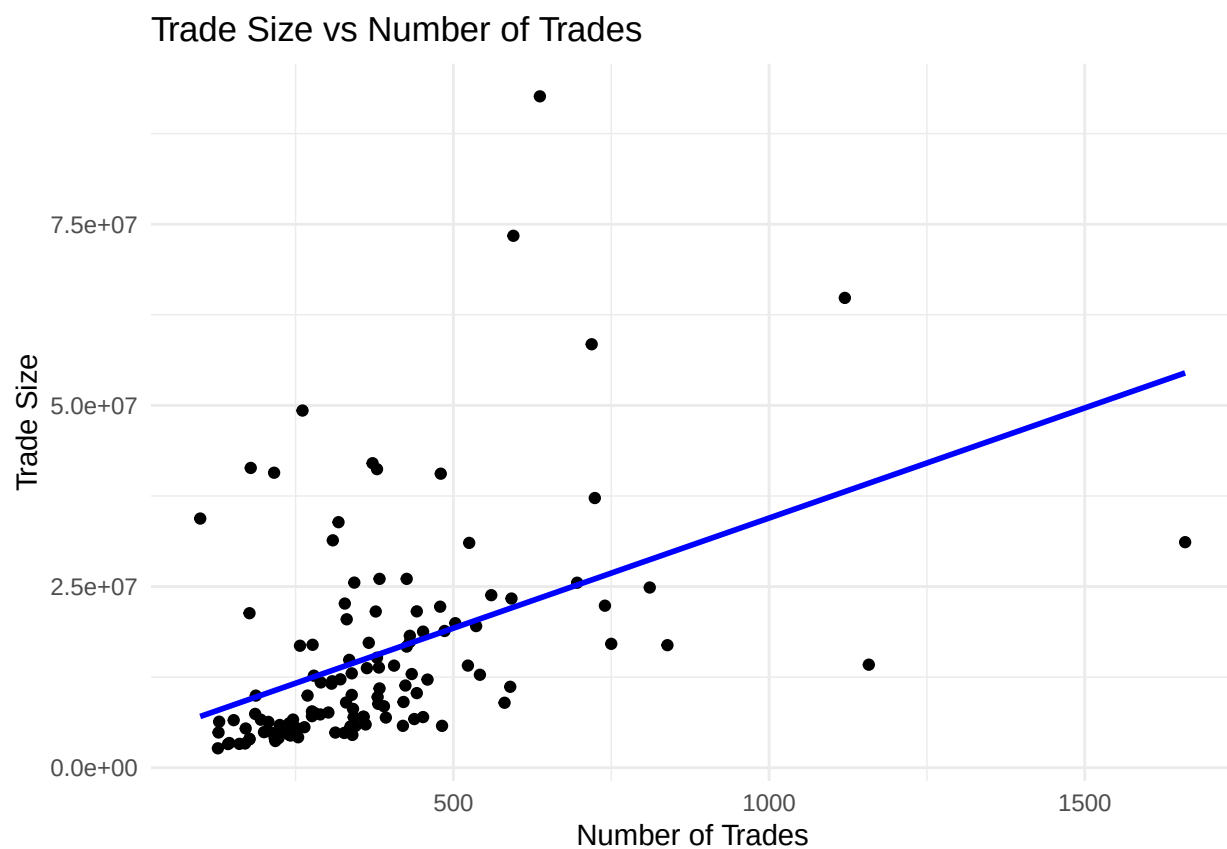
To assess whether a relationship exists between monthly trade volume and the number of transactions, both variables are plotted over time:





The plot raises some uncertainty about the relationship. Based on the analysis, the Spearman correlation between the number of trades per month and trade size per month is 0.6032014 and the Pearson correlation is 0.4444545, indicating a strong positive association.

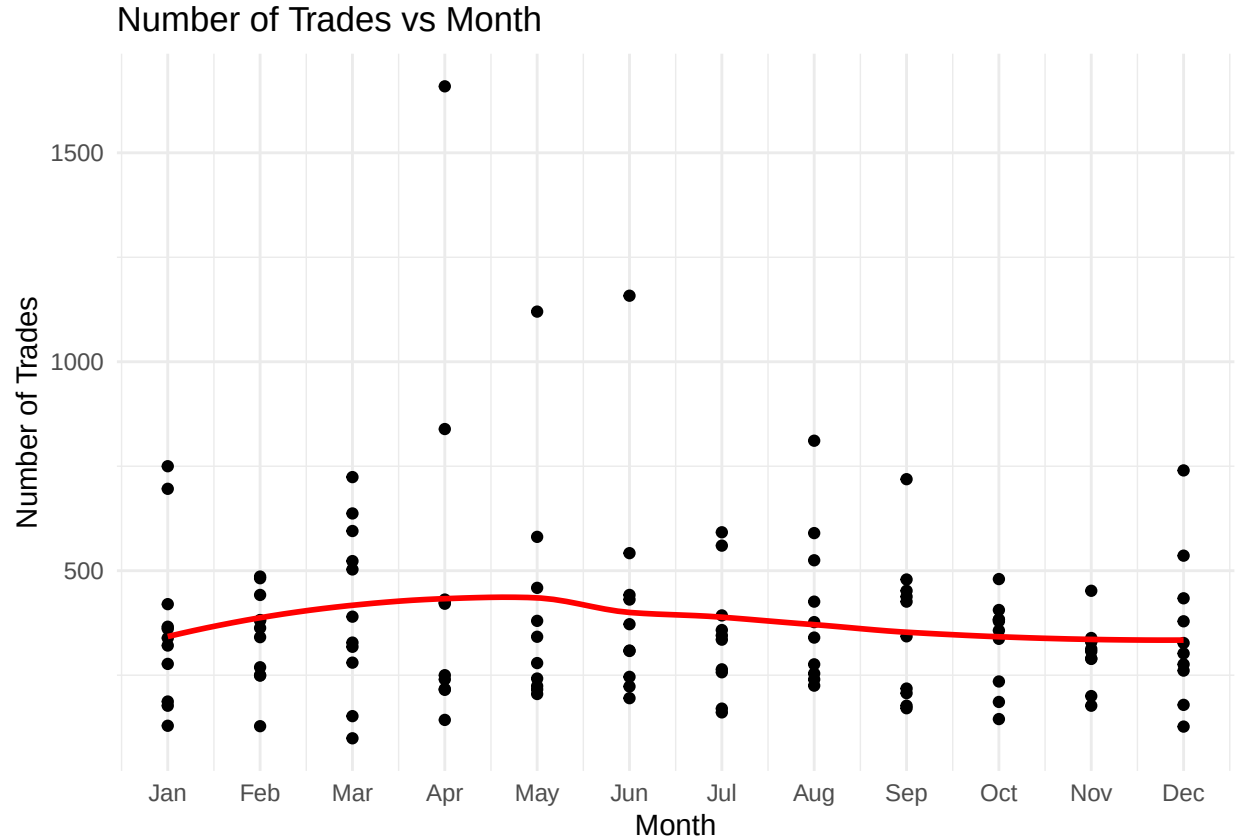
Overall, there is a strong correlation between monthly trade size and the number of trades. This is intuitive, as months with more transactions tend to exhibit higher total traded values. This relationship is further illustrated using a linear trend plot.



Next, we test whether the number of trades varies systematically across months or years.

The Spearman correlation per year is 0.0911164 and per month is -0.0642355, indicating no meaningful correlation.

The Spearman correlation between the number of trades and year, as well as between the number of trades and month, is close to zero. This indicates that trading activity does not exhibit significant seasonal or long-term time trends.



The corresponding plot confirms that the number of trades remains relatively stable across months. We can also compute the same for the trade size. The average trade size is 4.1135234×10^4

3) Chamber, party and excess return

3.1) Correlation and dominant party

We now analyze whether excess returns differ by political chamber or party.

The correlation between Chamber and excess return is -0.0128372, and between Chamber and net gain is -0.0151868.

For Party, the correlations are -0.0354317 and -0.0133452.

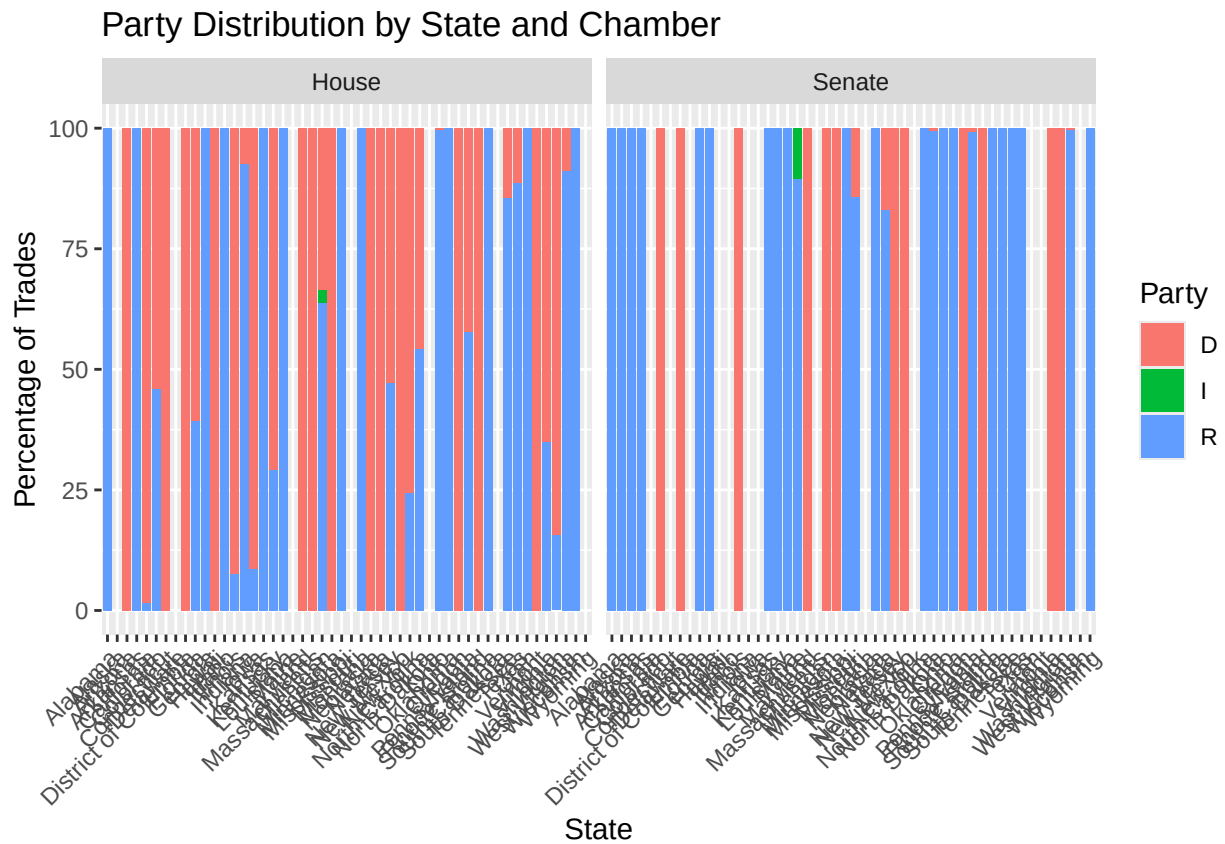
These results indicate very weak relationships between political chamber and excess return, as well as between party affiliation and excess return. Therefore, neither chamber nor party appears to be strongly associated with trading performance.

Next, trades are grouped by party and chamber within each state. The results are presented below.

```
## # A tibble: 6 x 4
##   State   Chamber Party Num_Trades
##   <chr>   <chr>   <chr>     <int>
## 1 Alabama House    R         371
## 2 Alabama Senate   R         976
## 3 Alaska  Senate   R         118
## 4 Arizona House    D         358
## 5 Arizona Senate   R          30
## 6 Arkansas House    R          45
```

Let us now examine the distribution of party affiliation across states.

```
## # A tibble: 6 x 4
##   State   Chamber Party Num_Trades
##   <chr>   <chr>   <chr>     <int>
## 1 Alabama House     R         371
## 2 Alabama Senate    R         976
## 3 Alaska  Senate    R         118
## 4 Arizona House     D        358
## 5 Arizona Senate    R          30
## 6 Arkansas House     R          45
```



Next, we identify the dominant political party within each state and chamber using both heatmaps and geographic maps.

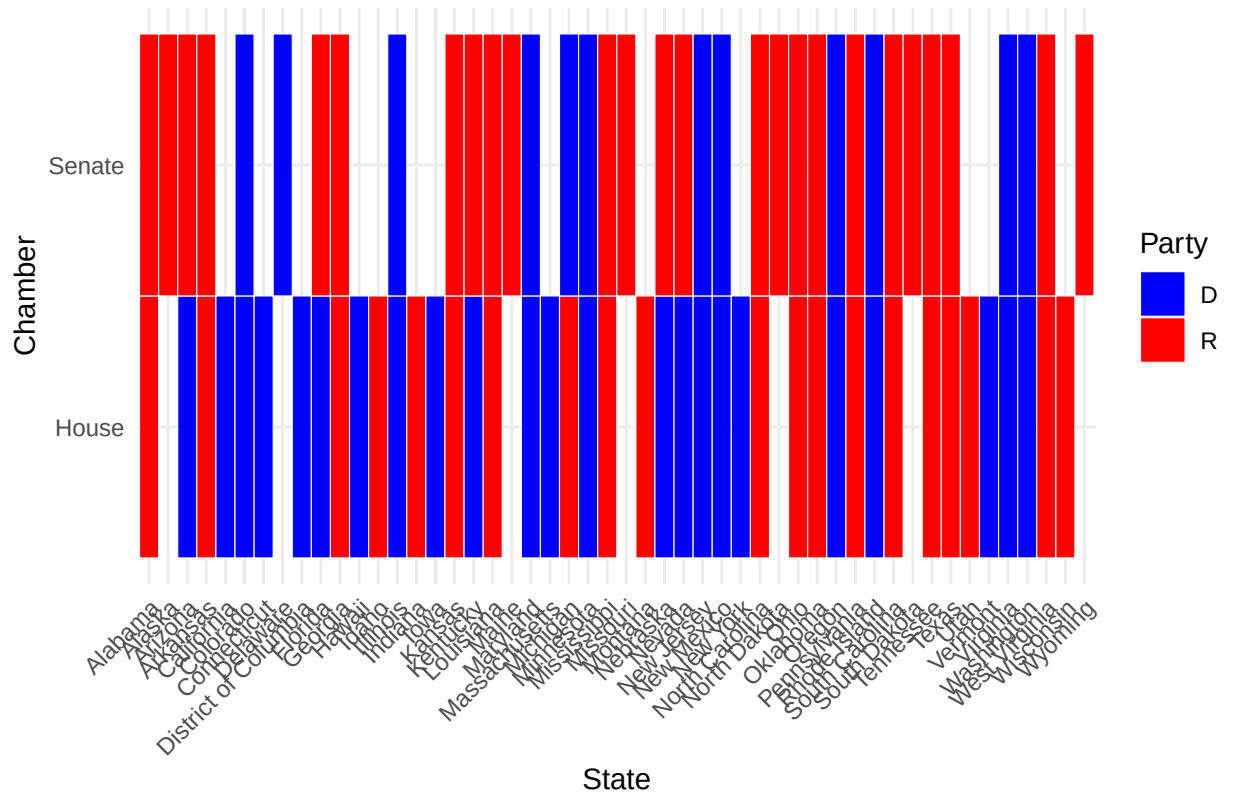
```
## # A tibble: 80 x 4
## # Groups:   State, Chamber [80]
##   State   Chamber Party Num_Trades
##   <chr>   <chr>   <chr>     <int>
## 1 Alabama House     R         371
## 2 Alabama Senate    R         976
## 3 Alaska  Senate    R         118
## 4 Arizona House     D        358
## 5 Arizona Senate    R          30
## 6 Arkansas House     R          45
## 7 Arkansas Senate    R          19
```

```
## 8 California House D 3588
## 9 Colorado House D 91
## 10 Colorado Senate D 88
## # i 70 more rows
```

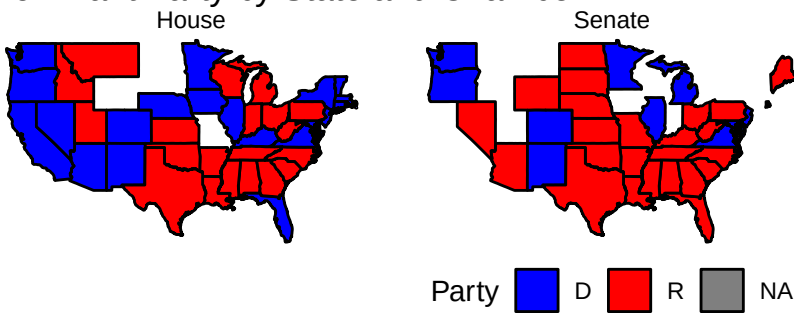
```
## # A tibble: 80 x 4
## # Groups:   State, Chamber [80]
##   State      Chamber Party Num_Trades
##   <chr>      <chr> <chr>    <int>
## 1 Alabama    House   R        371
## 2 Alabama    Senate  R        976
## 3 Alaska     Senate  R        118
## 4 Arizona    House   D        358
## 5 Arizona    Senate  R         30
## 6 Arkansas    House   R         45
## 7 Arkansas    Senate  R         19
## 8 California House   D       3588
## 9 Colorado    House   D         91
## 10 Colorado   Senate  D         88
## 11 Connecticut House   D        147
## 12 Delaware    Senate  D       751
## 13 District of Columbia House D         2
## 14 Florida     House   D      2358
## 15 Florida     Senate  R         6
## 16 Georgia     House   R       404
## 17 Georgia     Senate  R     2549
## 18 Hawaii      House   D         8
## 19 Idaho       House   R       140
## 20 Illinois    House   D       568
## 21 Illinois    Senate  D         22
## 22 Indiana     House   R       102
## 23 Iowa        House   D      203
## 24 Kansas      House   R       186
## 25 Kansas      Senate  R       641
## 26 Kentucky    House   D       366
## 27 Kentucky    Senate  R        28
## 28 Louisiana   House   R        19
## 29 Louisiana   Senate  R       165
## 30 Maine       Senate  R       349
## 31 Maryland    House   D         5
## 32 Maryland    Senate  D        10
## 33 Massachusetts House   D      747
## 34 Michigan    House   R      559
## 35 Michigan    Senate  D        57
## 36 Minnesota   House   D     842
## 37 Minnesota   Senate  D        28
## 38 Mississippi House   R        60
## 39 Mississippi Senate  R     529
## 40 Missouri    Senate  R         6
## 41 Montana     House   R     1331
## 42 Nebraska    House   D        31
## 43 Nebraska    Senate  R        29
## 44 Nevada      House   D     1273
## 45 Nevada      Senate  R        69
```

## 46 New Jersey	House	D	3209
## 47 New Jersey	Senate	D	13
## 48 New Mexico	House	D	1
## 49 New Mexico	Senate	D	2
## 50 New York	House	D	1928
## 51 North Carolina	House	R	930
## 52 North Carolina	Senate	R	98
## 53 North Dakota	Senate	R	184
## 54 Ohio	House	R	401
## 55 Ohio	Senate	R	1
## 56 Oklahoma	House	R	627
## 57 Oklahoma	Senate	R	365
## 58 Oregon	House	D	361
## 59 Oregon	Senate	D	240
## 60 Pennsylvania	House	R	213
## 61 Pennsylvania	Senate	R	150
## 62 Rhode Island	House	D	439
## 63 Rhode Island	Senate	D	887
## 64 South Carolina	House	R	76
## 65 South Carolina	Senate	R	5
## 66 South Dakota	Senate	R	1
## 67 Tennessee	House	R	621
## 68 Tennessee	Senate	R	958
## 69 Texas	House	R	3839
## 70 Texas	Senate	R	11
## 71 Utah	House	R	435
## 72 Vermont	House	D	102
## 73 Virginia	House	D	752
## 74 Virginia	Senate	D	11
## 75 Washington	House	D	315
## 76 Washington	Senate	D	167
## 77 West Virginia	House	R	564
## 78 West Virginia	Senate	R	442
## 79 Wisconsin	House	R	47
## 80 Wyoming	Senate	R	3

Dominant Party by State and Chamber



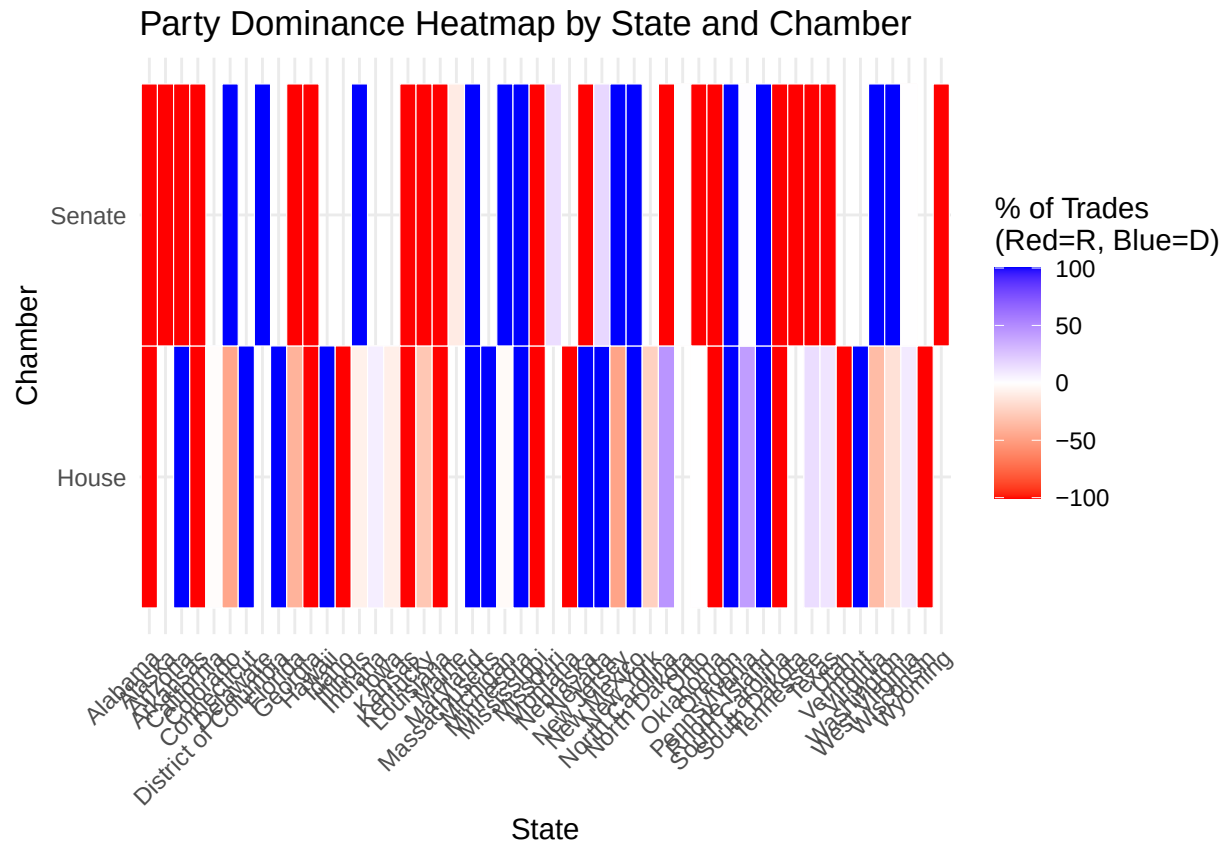
Dominant Party by State and Chamber



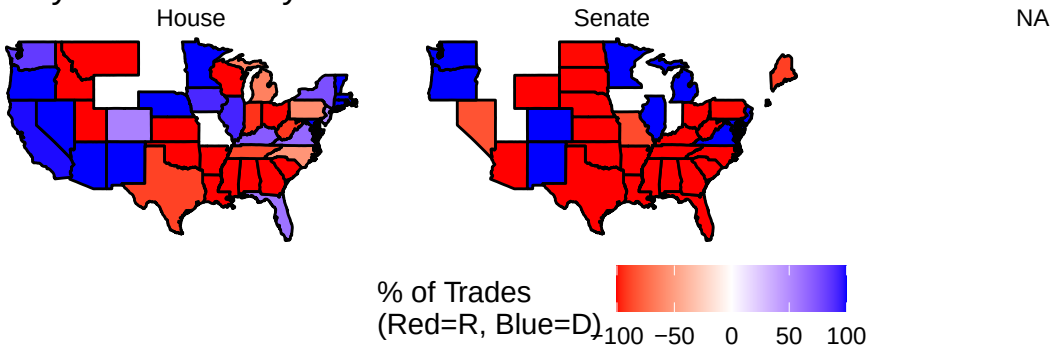
NA

4

We now examine the intensity of party dominance across states.



Party Dominance by State and Chamber



3.2) Excess return by Party

```
## # A tibble: 3 x 5
##   Party Mean_Excess_Return Median_Excess_Return SD_Excess_Return Num_Trades
##   <chr>          <dbl>          <dbl>          <dbl>      <int>
## 1 D             -8.34           -22.6           207.       21882
## 2 I            -34.6           -52.2           102.         64
## 3 R            -21.2           -33.0           150.      24521
```

We now examine the statistical properties of excess returns.

```
##   Min.  1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
## -318.75 -84.40  -27.71  -15.03  16.10 10885.41  2506
```

The initial summary appears highly negative, which is primarily due to the presence of missing values. After excluding missing observations, the corrected statistics are as follows:

The mean excess return is: -15.0291822 The median excess return is: -27.7138947 The standard deviation is: 179.90644

Interpretation

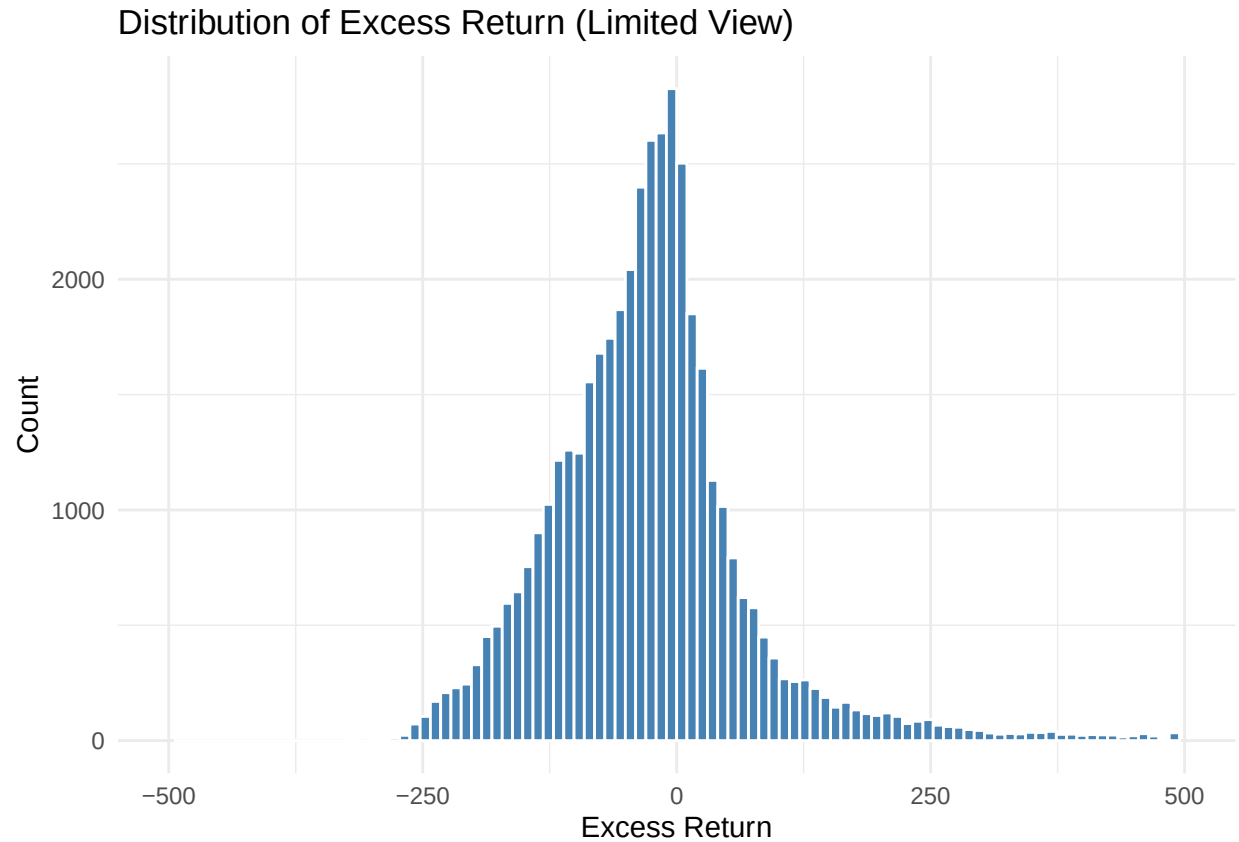
On average, trades underperformed the market benchmark by approximately 15%.

The mean is slightly higher than the median because a small number of large positive outcomes increase the average.

More than half of all trades underperformed the benchmark by roughly 28%, confirming a negatively skewed distribution.

The very high standard deviation indicates substantial volatility in trade outcomes, with extreme outliers such as 10,885 and −318 strongly influencing dispersion.

Distribution of excess returns - Histogram



The histogram suggests an approximately normal distribution with a pronounced right tail, indicating positive skewness driven by extreme high returns.

A negative value of `net_gain` represents a loss, while a positive value represents a gain. The top ten overperforming trades with highest returns based on net gain, are shown below.

##	Name	Party	Company	Ticker	net_gain
## 1	Whitehouse, Sheldon	D	NVIDIA Corporation	NVDA	10885.406
## 2	Whitehouse, Sheldon	D	NVIDIA Corporation	NVDA	10427.902
## 3	Whitehouse, Sheldon	D	NVIDIA Corporation	NVDA	9432.152
## 4	Whitehouse, Sheldon	D	NVIDIA Corporation	NVDA	9385.926
## 5	Whitehouse, Sheldon	D	NVIDIA Corporation	NVDA	7852.900
## 6	Brenda Lulenar Lawrence	D	NVIDIA CORPORATION	NVDA	3334.196
## 7	Debbie Dingell	D	NVIDIA CORPORATION	NVDA	3169.894
## 8	Whitehouse, Sheldon	D	NVIDIA Corporation	NVDA	3115.648
## 9	Pete Sessions	R	NVIDIA CORPORATION	NVDA	3058.806
## 10	Whitehouse, Sheldon	D	NVIDIA Corporation	NVDA	2422.490

The most frequently traded ticker is MSFT.

Individual-level aggregation:

Trades were aggregated by individual politician to compute total activity and outcomes.

```
## # A tibble: 341 x 10
##   Name      Party State Chamber total_trades total_trade_value tot_excess_return
##   <chr>    <chr> <chr> <chr>         <int>         <dbl>         <dbl>
## 1 Thomas ~ R    New ~ House         2758         85266879        -118473.
## 2 Josh Go~ D    New ~ House         2574         264384287         21352.
## 3 Perdue,~ R    Geor~ Senate        2242         49400621         -59108.
## 4 Lois Fr~ D    Flor~ House         1358         11060679         -38700.
## 5 Greg Gi~ R    Mont~ House         1331         98100668.         -27173.
## 6 Susie L~ D    Neva~ House         1261         10440131          -9388.
## 7 Alan S.~ D    Cali~ House         1243         11299134.         -18951.
## 8 Daniel ~ D    New ~ House         1184         32031592.         -17808.
## 9 Gilbert~ D    Cali~ House         1142         12725071          1733.
## 10 K. Mich~ R    Texas House          977         16829488.         -21756.
## # i 331 more rows
## # i 3 more variables: Number_of_sale <int>, Number_of_purchase <int>,
## #   Net_gain <dbl>
```

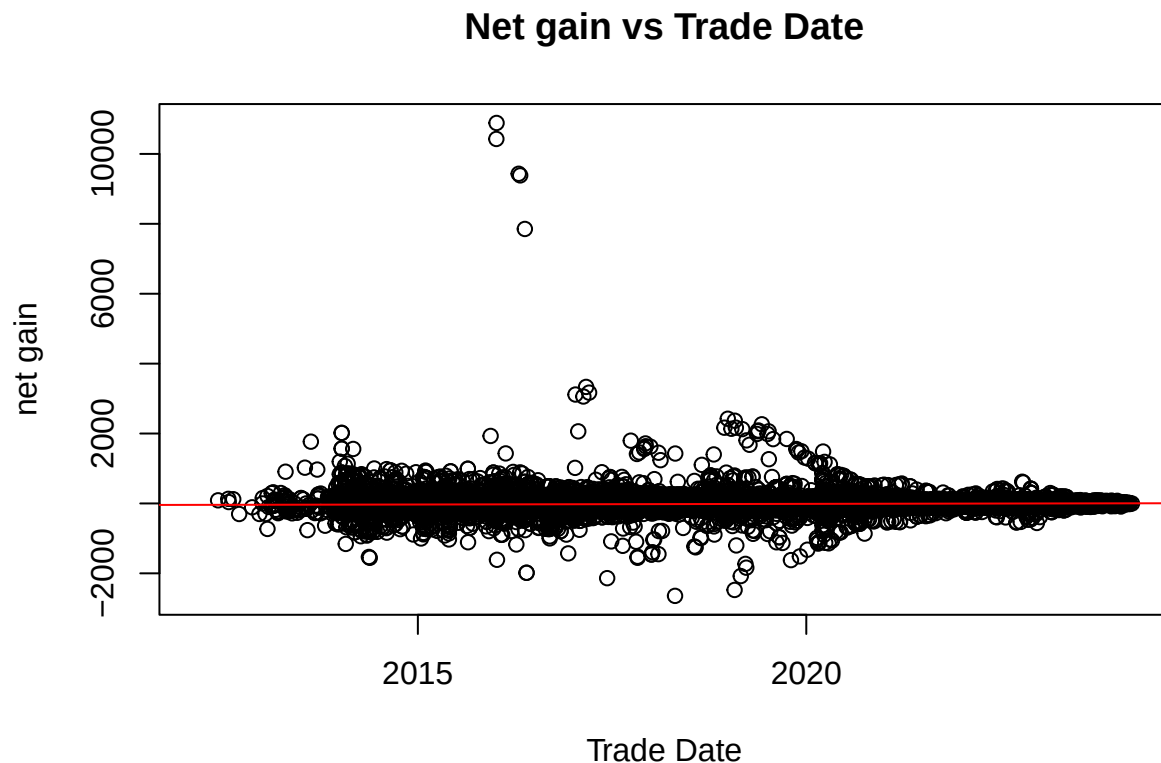
3.3) Time and performance

We test whether excess return is related to the timing of trades.

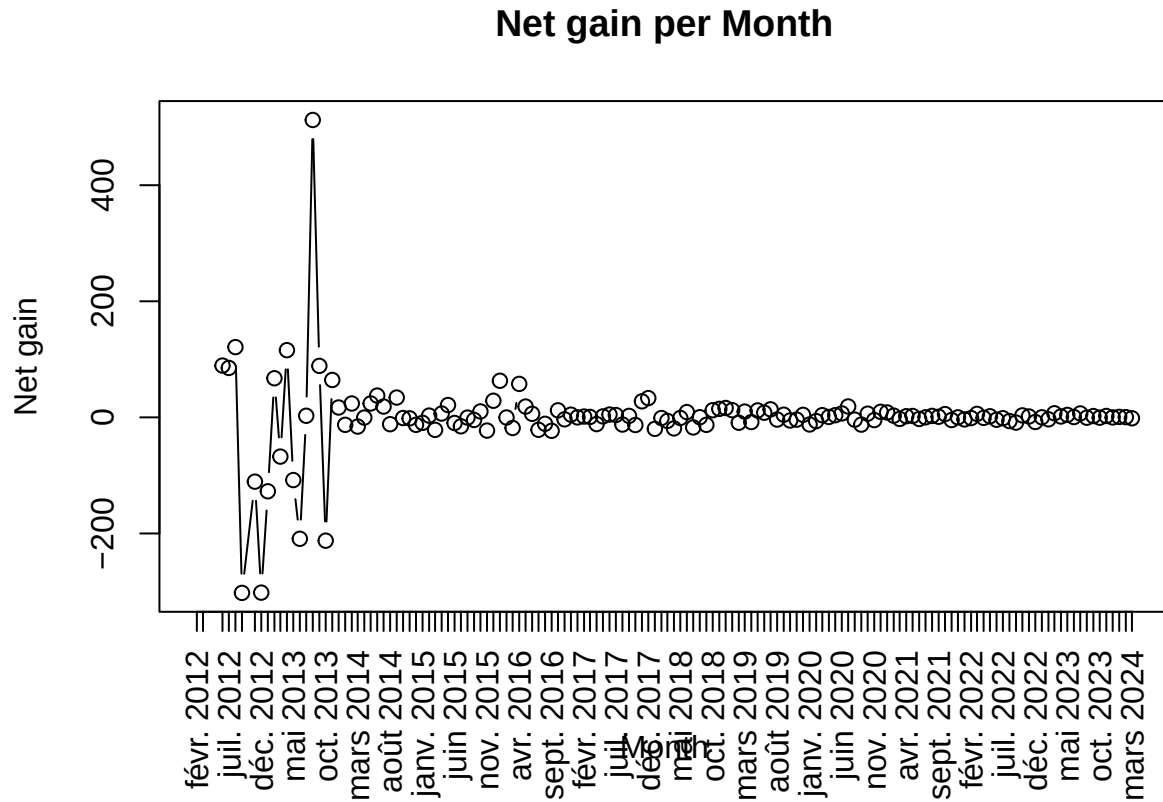
```
## [1] 0.05644884
```

The correlation is close to zero, indicating no direct relationship between trade date and excess return.

We further explore monthly and yearly patterns.



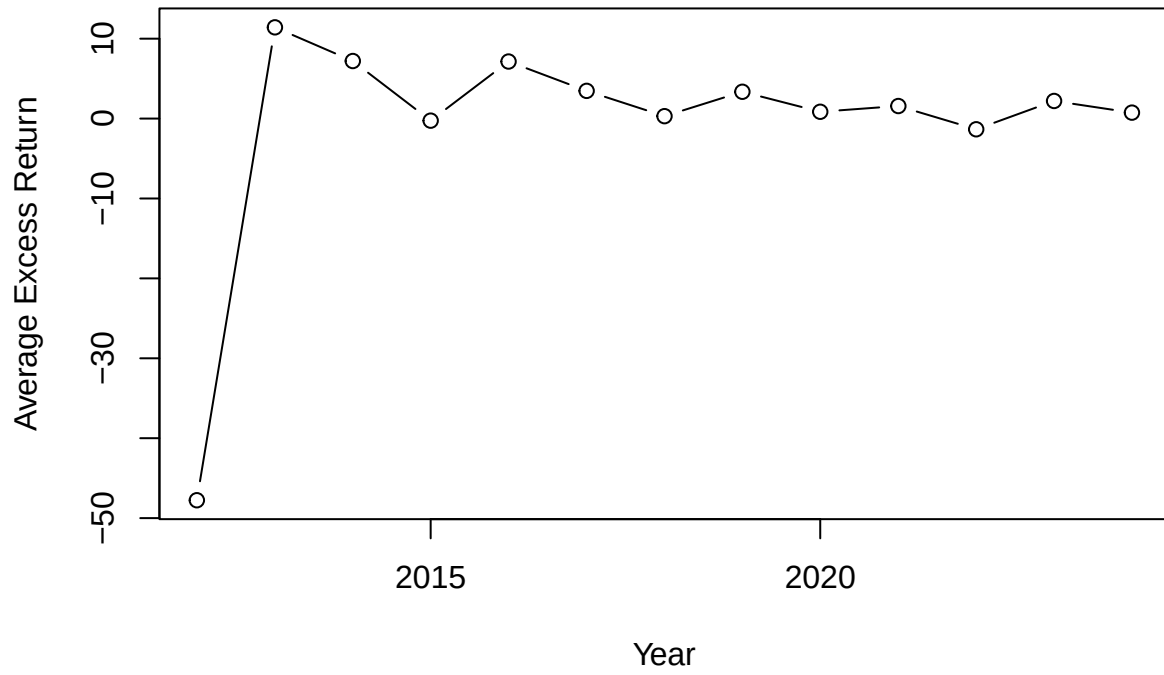
What is the relation between net gain and month?



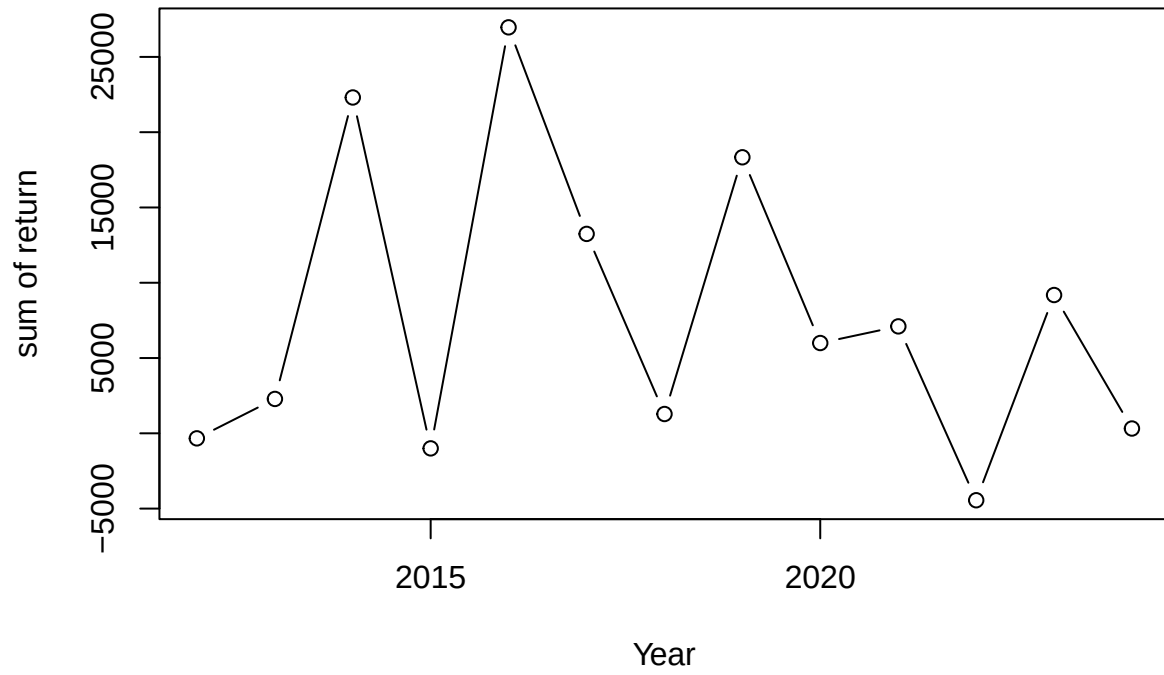
The average net gain shows a gradual increase over time but remains close to zero. Early years, particularly 2012, contain limited observations and should be interpreted with caution.

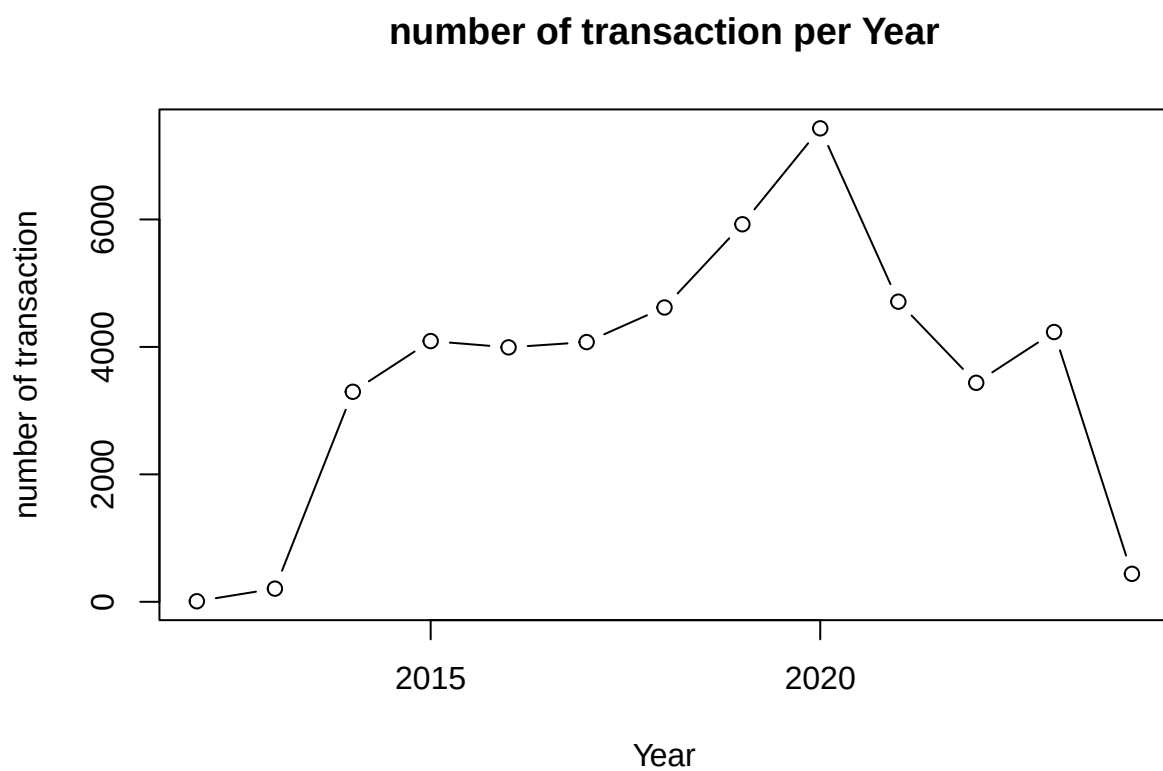
relation between excess of return and month

Average Excess Return per Year

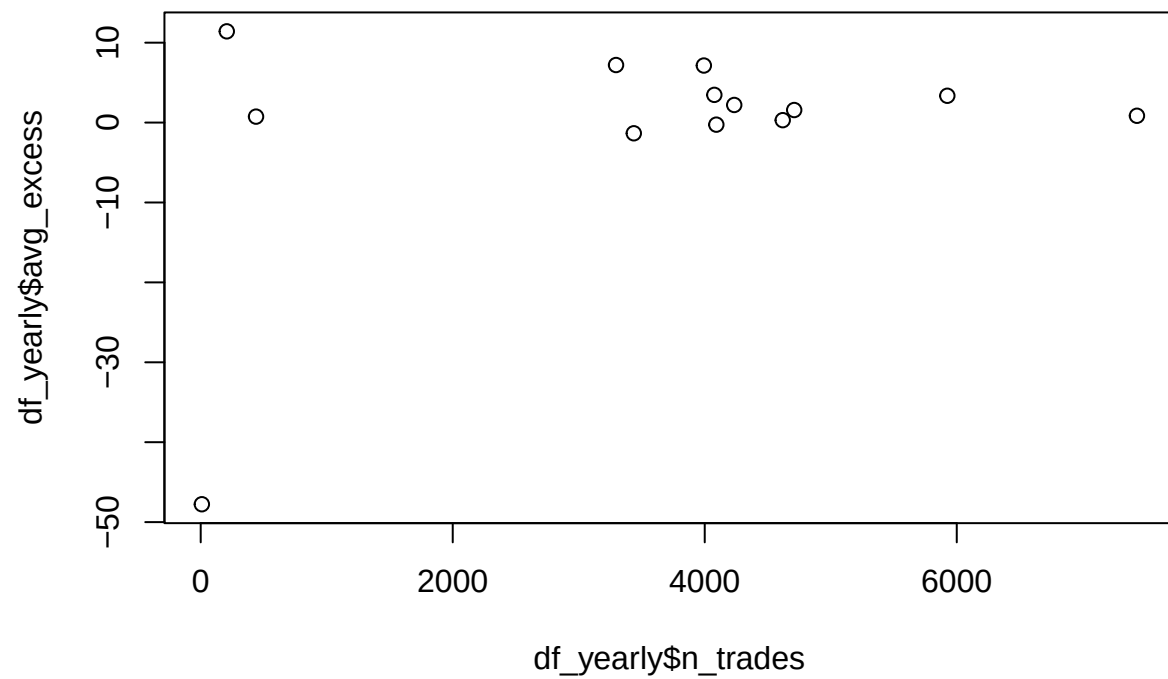


sum of excee return per Year





The average excess return increases over time. To avoid distortion from the low-activity year 2012, this year is excluded from further correlation analysis.

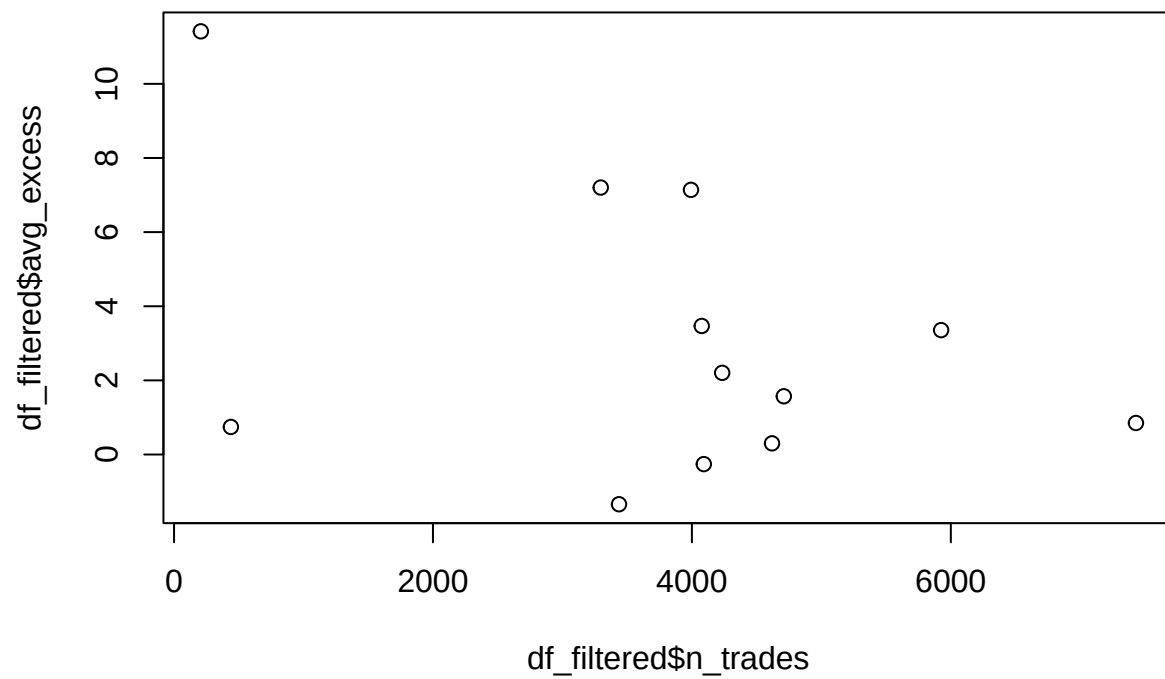


The point corresponding to 2012 biases the relationship.

The correlation is as follows:

```
## [1] -0.4131439
```

A modest positive relationship is observed between the number of trades and average net gain.

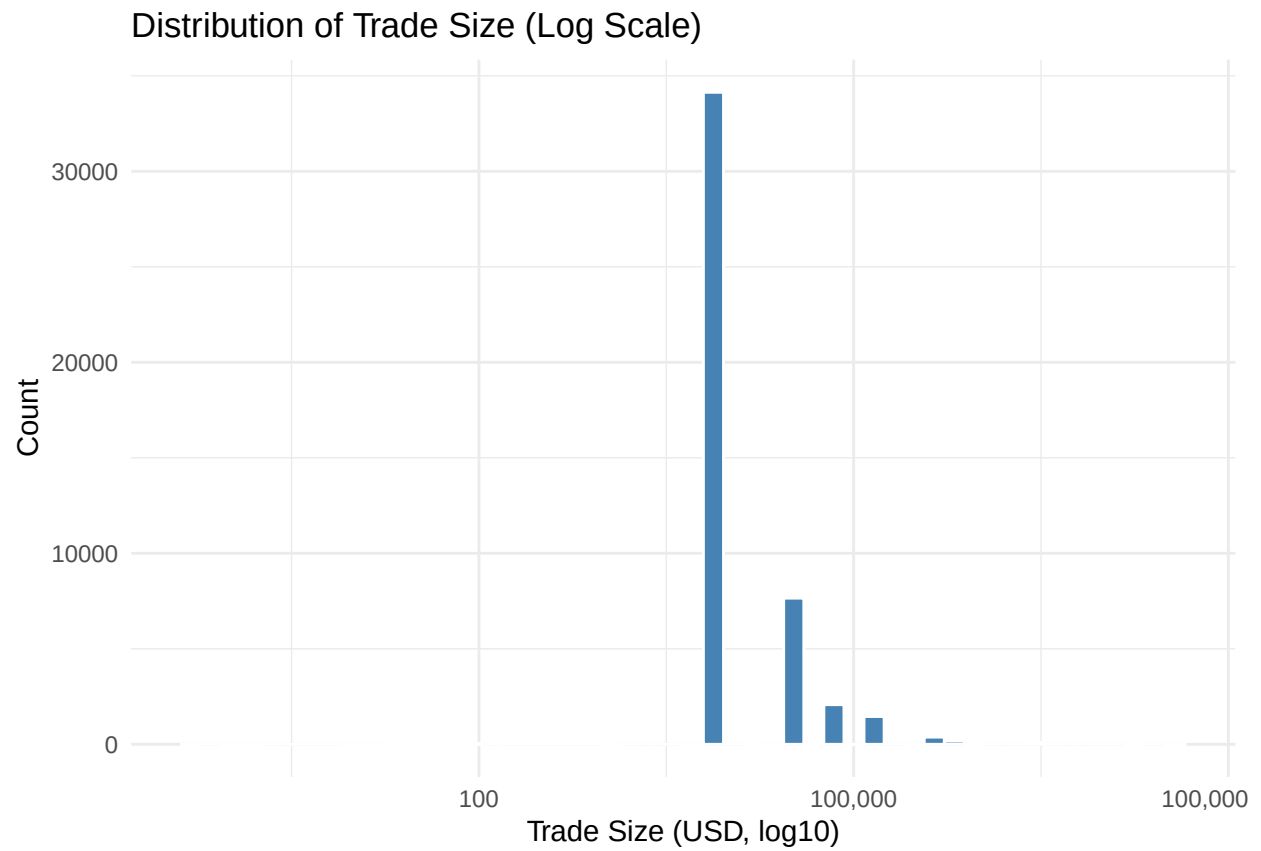


3.4) excess return, trade size by party

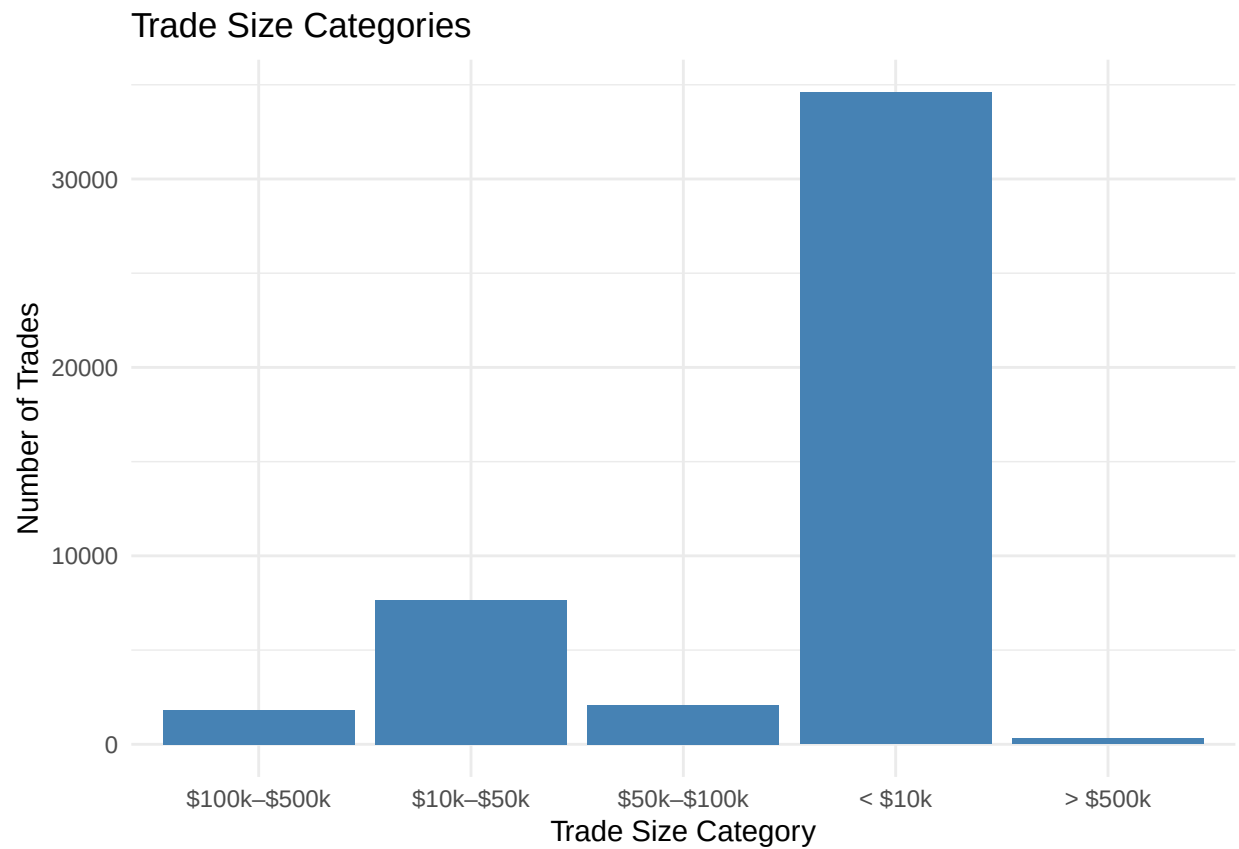
##1. Histogram of trade size

We begin by examining the distribution of trade size. Because trade sizes are highly skewed, we visualize the distribution on a logarithmic scale.

#LOG scale

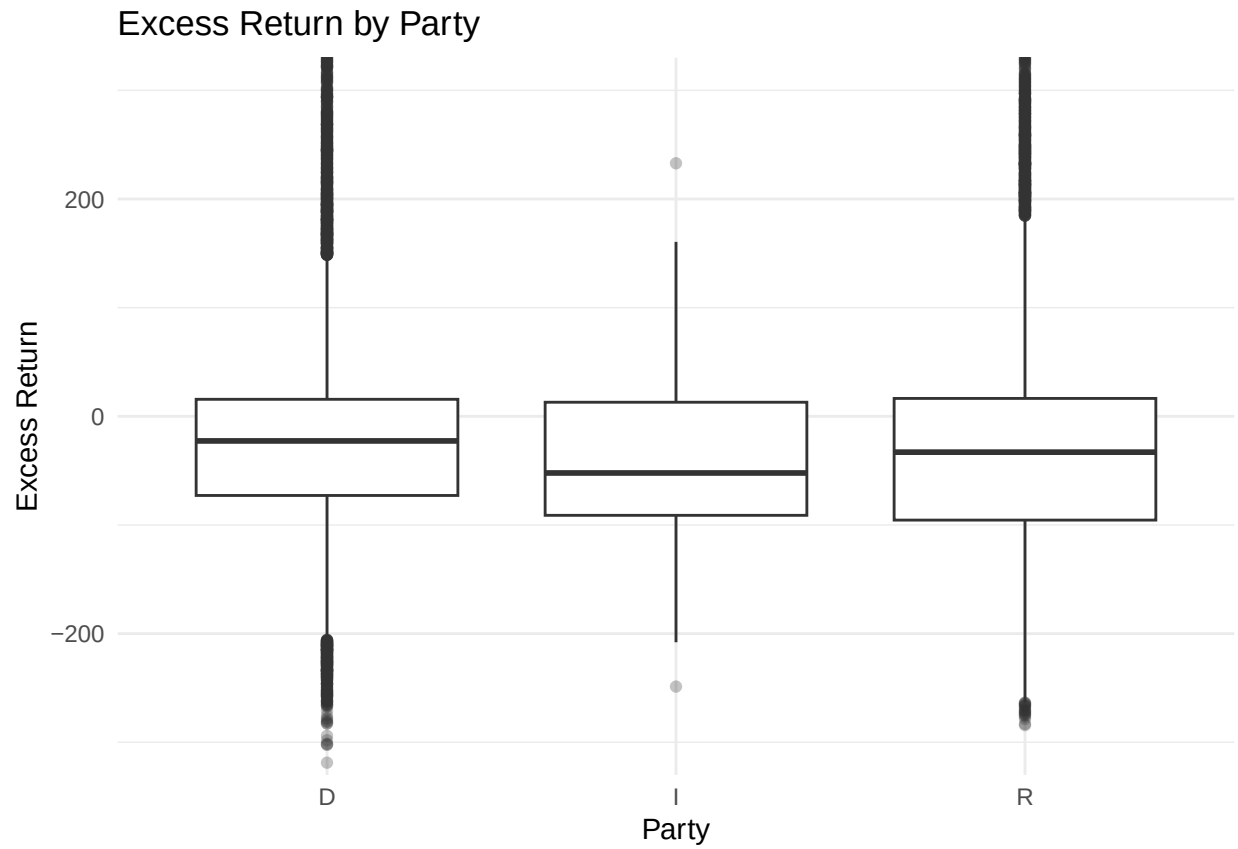


To improve interpretability, we also group trades into size categories.



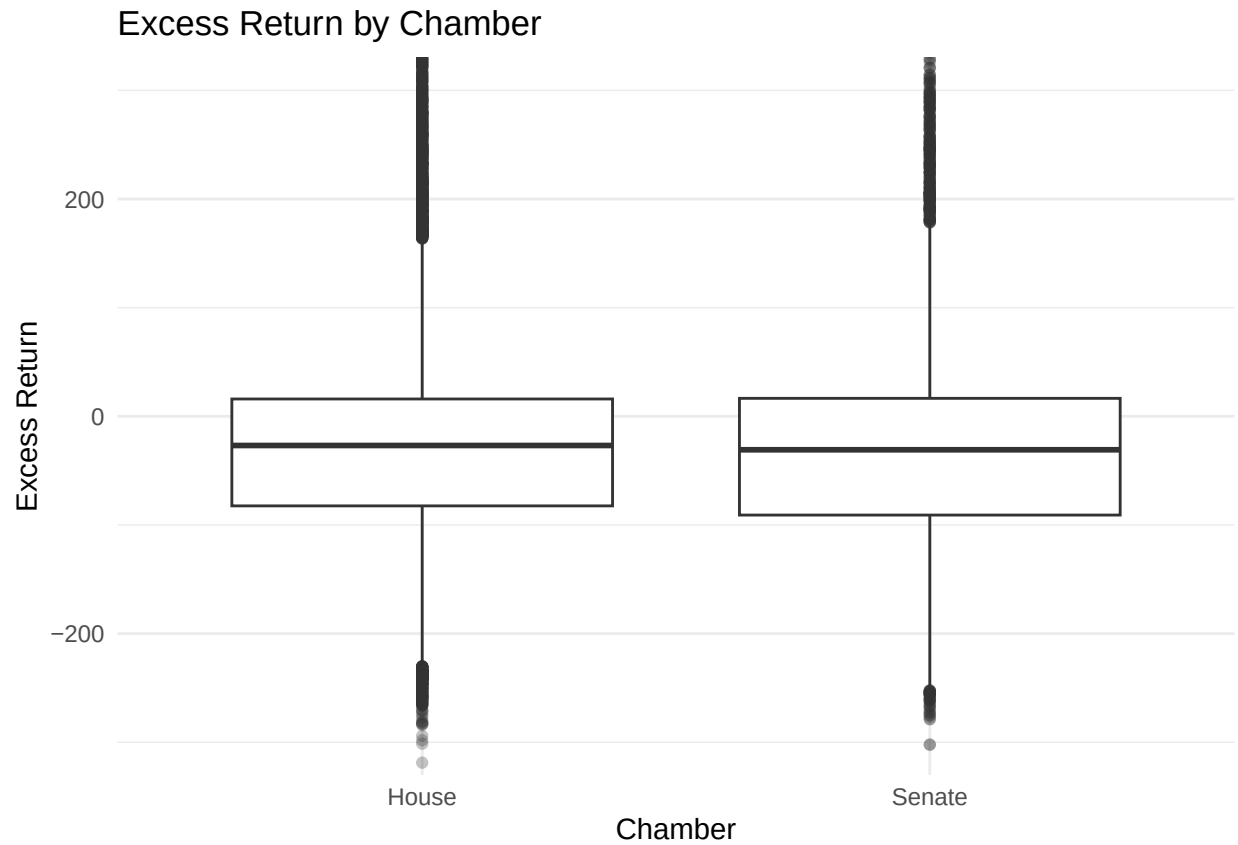
##2. Boxplot: excess return by party

We next compare the distribution of excess returns across political parties.



##3.1 Boxplot: excess return by chamber

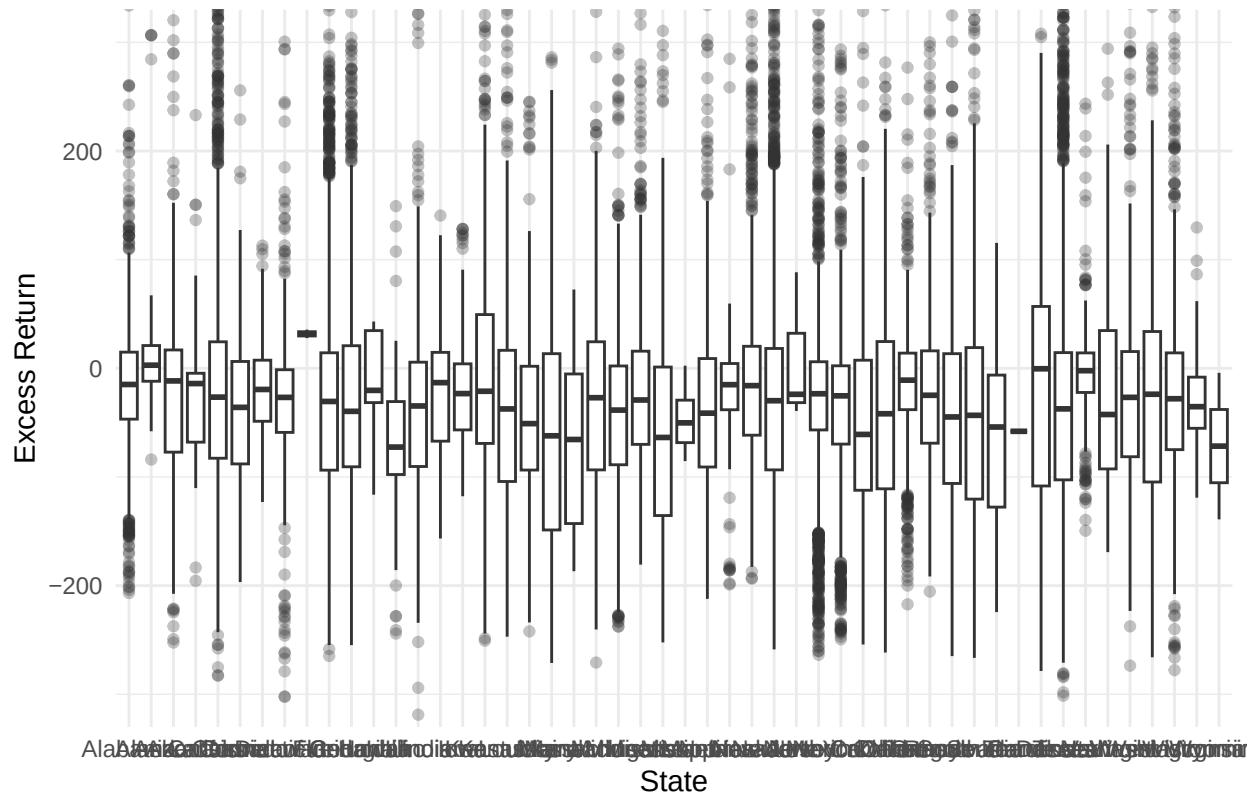
We also examine whether excess returns differ between the House and the Senate.



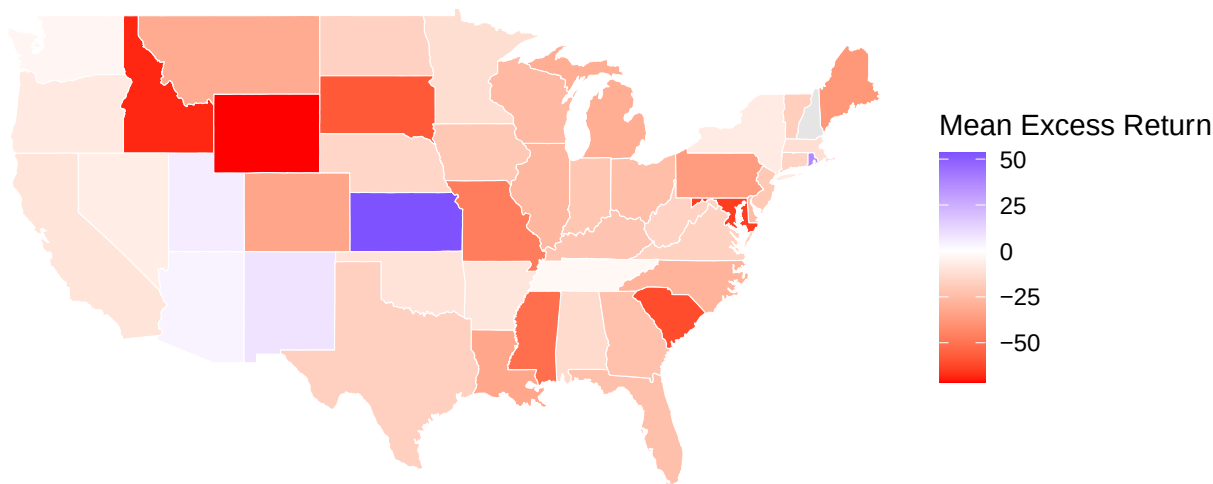
##3.2 excess return by state/ net gain by state

To explore geographic variation, we compare excess return distributions across states and compute state-level averages for both excess return and net gain.

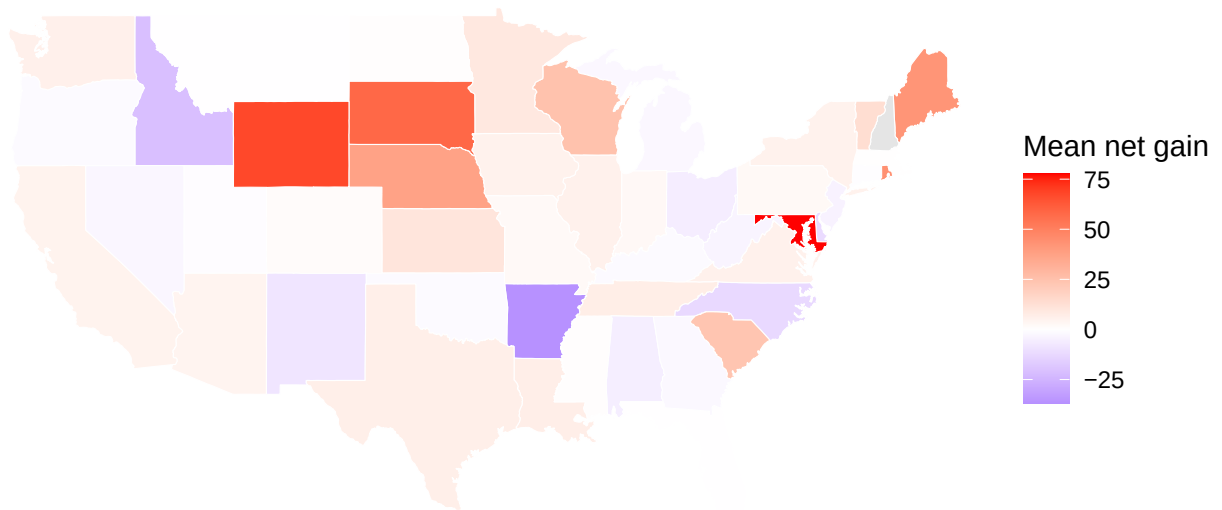
Excess Return by State



Mean Excess Return by State

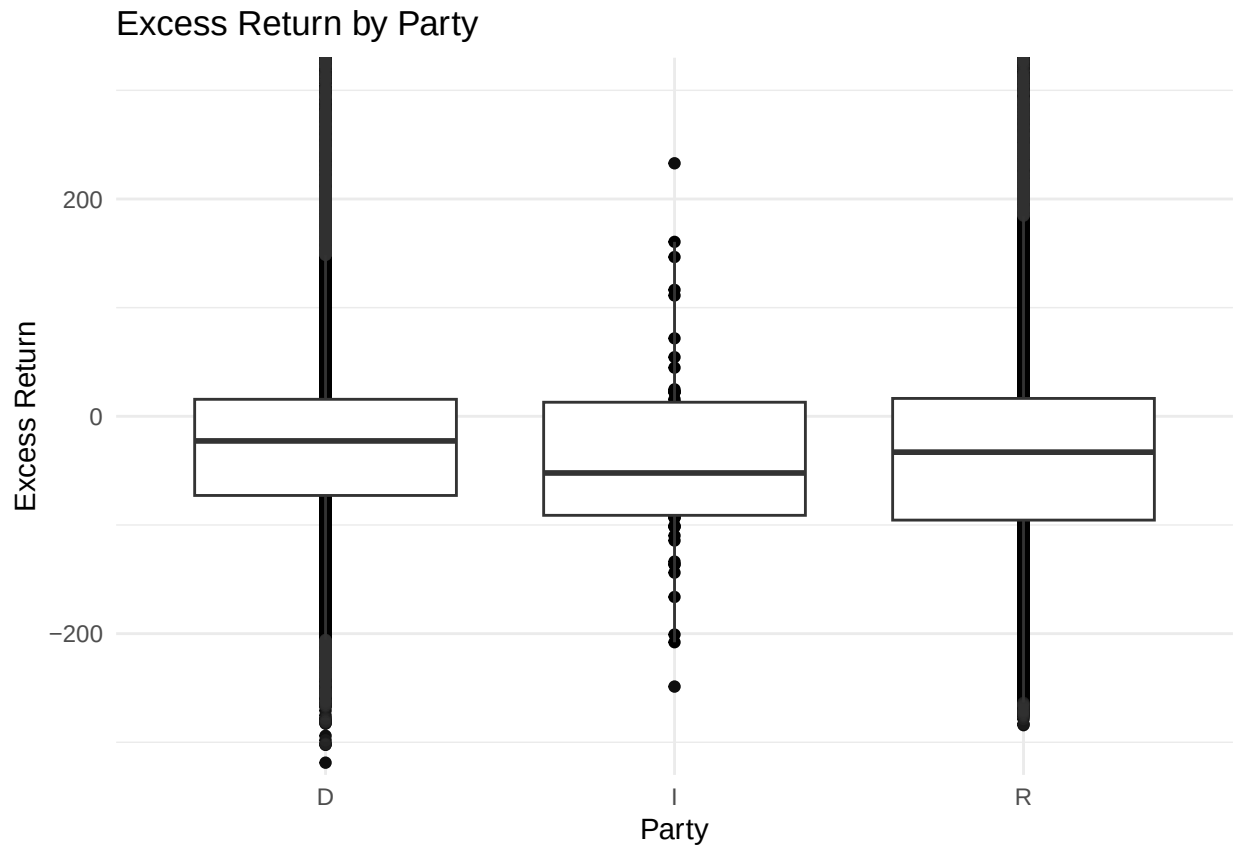


net gain by state



##3.3 Boxplot: excess return by party

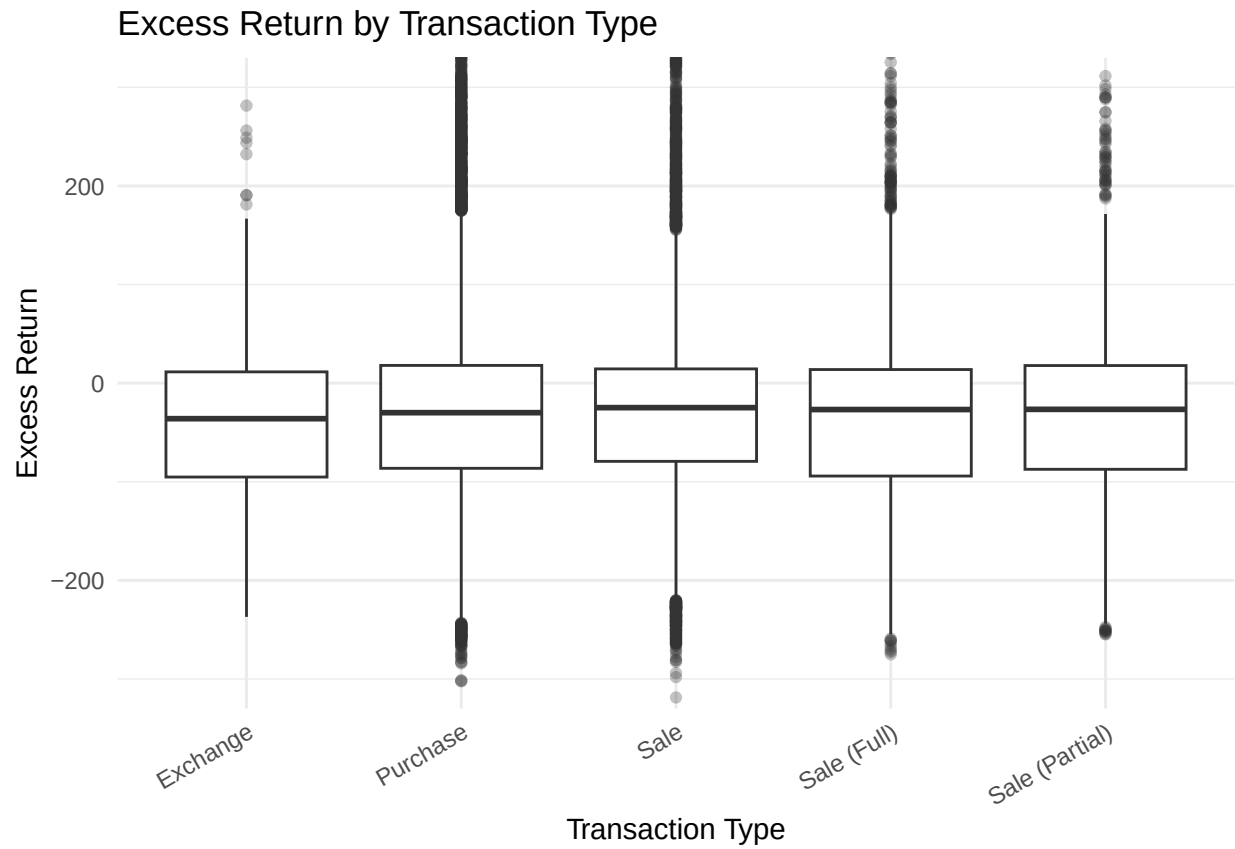
To visualize the dispersion and individual observations simultaneously, we overlay points on the party-level



boxplots.

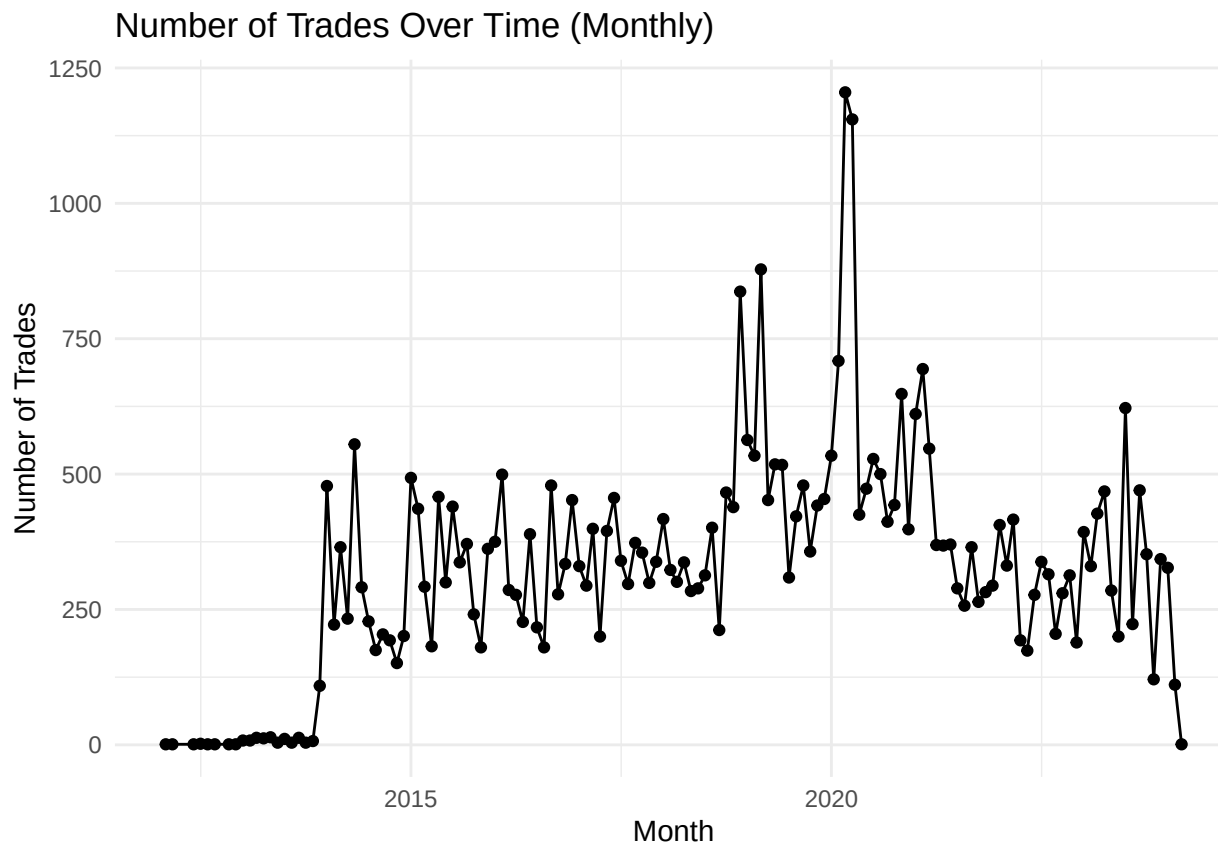
##4. Boxplot: excess return by transaction type

We further compare excess returns across transaction types (e.g., purchases and sales).



##6. Clean time series: number of trades over time

Finally, we construct a monthly time series of the number of trades to illustrate how trading activity evolves over time.



Correlation between trade size and party

We test whether total traded value differs between parties using a two-sample t-test.

```
##
## Welch Two Sample t-test
##
## data: total_trade_value by Party
## t = 0.22042, df = 242.48, p-value = 0.8257
## alternative hypothesis: true difference in means between group D and group R is not equal to 0
## 95 percent confidence interval:
## -4255936 5328439
## sample estimates:
## mean in group D mean in group R
##      5938708      5402457

## function (x, ...)
## UseMethod("t.test")
## <bytecode: 0x00000253bfe781c0>
## <environment: namespace:stats>
```

The test suggests no statistically significant difference between the two parties; trade values appear broadly balanced.

3.5) Sales and purchases

We summarise total traded value by transaction type and report the number of sales, purchases, and exchanges.

```
## # A tibble: 5 x 2
##   Transaction    Trade_size
##   <chr>          <dbl>
## 1 Exchange      20985670.
## 2 Purchase      937178784.
## 3 Sale          719224917.
## 4 Sale (Full)   112482538.
## 5 Sale (Partial) 121558986.
```

The number of sale is 23045 The number of purchase is 23013 The number of exchanges is 409

A correlation comparison between sales and purchases is not meaningful because the number of observations differs substantially between these transaction types.

#Conclusion

This study finds no significant relationship between excess returns and political party affiliation or chamber membership. Correlation analysis and group comparisons consistently show that trading performance does not systematically differ across political groups.

Despite the relatively large dataset, trading activity is unevenly distributed across time, states, and individuals. To improve economic interpretation, excess returns were transformed into net gains using trade direction. The analysis highlights strong patterns in party and chamber dominance across states, as well as a clear relationship between trading volume and the number of transactions. However, performance outcomes appear to be driven primarily by individual behavior rather than political characteristics.

#Limitations

Several limitations affect the precision of the results. The dataset lacks matched buy–sell pairs, preventing a direct measurement of realized profits. In addition, missing values and uneven data coverage across years and regions introduce potential bias. Finally, the limited geographic detail restricts a full assessment of regional effects on excess returns.

#Use of AI

The use of AI was necessary to resolve several technical issues encountered during the project. For example, it was used to address knitting problems related to LaTeX compilation through the configuration of the `latex_engine`, such as `xelatex`.

In addition, AI assisted in identifying and applying appropriate R functions and packages for data processing and visualization, including tools such as `lubridate` and `maps`. Many code segments were initially generated with the help of AI; however, they were always reviewed, modified, and fully understood by the authors, particularly for data cleaning and reorganization tasks.

AI was used primarily as a technical support tool for coding rather than for generating analytical ideas or research direction. The R Markdown structure and overall report organization were created independently without the use of AI.

#task distribution

Julien :

- 1) Organization of the data : ## Converting trade size into interval ## Creating Midpoint for Trade Size ## calculate the net gain/lost
- 2) Trade Size and number
- 3) Chamber, party and excess return 3.1) Correlation and dominant party (only maps of state) 3.3) Time and performance 3.5) Sales and purchases

Rmarkdown part 1,2,3.1,3.2

Lukas :

- 1) Organization of the data : ## cleaning data all across all column by converting dates to numbers ## delete tickertype ## Creating Trade Delay (in days) ## Creating Net Trade Direction
- 2) Trade Size and number (interpretation)
- 3) Chamber, party and excess return 3.1) Correlation and dominant party 3.2) Excess return by Party 3.4) excess return, trade size by party

Rmarkdown part 3.3, 3.4, 3.5